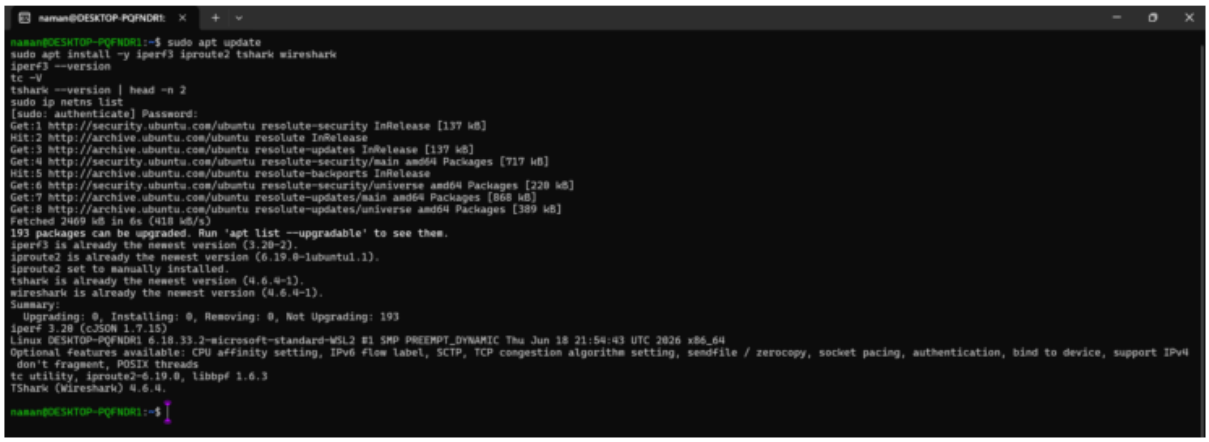
CNNP SKILL 3

Name: Naman Masineni Roll Number: 2520090194

Tool installation and version verification:



Network namespaces, IP configuration, routing and ping:

```
naman@DESKTOP-PQFNDR1:~$ # Create three network namespaces
sudo ip netns add sender
sudo ip netns add router
sudo ip netns add receiver

# Create virtual network cables
sudo ip link add veth-s type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-d

# Connect cables to namespaces
sudo ip link set veth-s netns sender
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-d netns receiver

# Assign IP addresses
sudo ip netns exec sender ip addr add 192.168.10.2/24 dev veth-s
sudo ip netns exec router ip addr add 192.168.10.1/24 dev veth-r1
sudo ip netns exec router ip addr add 192.168.20.1/24 dev veth-r2
sudo ip netns exec receiver ip addr add 192.168.20.2/24 dev veth-d

# Enable interfaces
sudo ip netns exec sender ip link set veth-s up
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec receiver ip link set veth-d up

# Enable loopback
sudo ip netns exec sender ip link set lo up
sudo ip netns exec router ip link set lo up
sudo ip netns exec receiver ip link set lo up

# Enable routing
sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1

# Add routes
sudo ip netns exec sender ip route add 192.168.20.0/24 via 192.168.10.1
sudo ip netns exec receiver ip route add 192.168.10.0/24 via 192.168.20.1

# Check connectivity
sudo ip netns exec sender ping -c 4 192.168.20.2

# Display namespaces
sudo ip netns list
net.ipv4.ip_forward = 1
PING 192.168.20.2 (192.168.20.2) 56(84) bytes of data.
64 bytes from 192.168.20.2: icmp_seq=1 ttl=63 time=0.341 ms
64 bytes from 192.168.20.2: icmp_seq=2 ttl=63 time=0.341 ms
64 bytes from 192.168.20.2: icmp_seq=3 ttl=63 time=0.117 ms
64 bytes from 192.168.20.2: icmp_seq=4 ttl=63 time=0.095 ms

--- 192.168.20.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3073ms
rtt min/avg/max/mdev = 0.095/0.223/0.341/0.117 ms
receiver (id: 2)
router (id: 1)
sender (id: 0)
naman@DESKTOP-PQFNDR1:~$
```

Configure 5 Mbps router bottleneck and queue:

```
sender (id: 0)
naman@DESKTOP-PQFNDR1:~$ # Configure router bandwidth limit
sudo ip netns exec router tc qdisc replace dev veth-r2 root handle 1: htb default 10

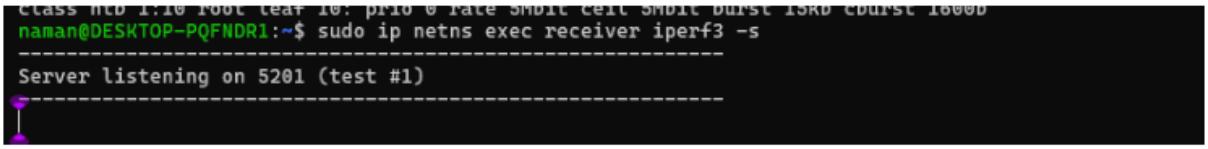
sudo ip netns exec router tc class replace dev veth-r2 parent 1: classid 1:10 htb rate 5mbit ceil 5mbit burst 15k

# Add a queue for packets
sudo ip netns exec router tc qdisc replace dev veth-r2 parent 1:10 handle 10: pfifo limit 100

# Verify the configuration
sudo ip netns exec router tc -s qdisc show dev veth-r2

sudo ip netns exec router tc class show dev veth-r2
qdisc htb 1: root refcnt 13 r2q 10 default 0x10 direct_packets_stat 0 direct_qlen 1000
 Sent 0 bytes 0 pkt (dropped 0, overlimits 0 requeues 0)
 backlog 0b 0p requeues 0
qdisc pfifo 10: parent 1:10 limit 100p
 Sent 0 bytes 0 pkt (dropped 0, overlimits 0 requeues 0)
 backlog 0b 0p requeues 0
class htb 1:10 root leaf 10: prio 0 rate 5Mbit ceil 5Mbit burst 15Kb cburst 1600b
naman@DESKTOP-PQFNDR1:~$
```

Start receiver iperf3 server:



Bandwidth test commands for 1–10 Mbps:

```
naman@DESKTOP-PQFNDR1:~$ mkdir -p ~/skill3_results

for rate in 1 2 4 5 7 10; do
    echo "========== TESTING ${rate} Mbps =========="
    sudo ip netns exec sender iperf3 \
      -c 192.168.20.2 \
      -u -b ${rate}M -t 10 \
      --get-server-output \
      | tee ~/skill3_results/rate_${rate}Mbps.txt
    sleep 2
done

echo "========== ROUTER QUEUE STATISTICS =========="
sudo ip netns exec router tc -s qdisc show dev veth-r2
========== TESTING 1 Mbps ==========
[sudo: authenticate] Password:
```

Bandwidth tests: 1 and 2 Mbps:

```
naman@DESKTOP-PQFNDR1:~$ mkdir -p ~/skill3_results

for rate in 1 2 4 5 7 10; do
    echo "========== TESTING ${rate} Mbps =========="
    sudo ip netns exec sender iperf3 \
      -c 192.168.20.2 \
      -u -b ${rate}M -t 10 \
      --get-server-output \
      | tee ~/skill3_results/rate_${rate}Mbps.txt
    sleep 2
done

echo "========== ROUTER QUEUE STATISTICS =========="
sudo ip netns exec router tc -s qdisc show dev veth-r2
========== TESTING 1 Mbps ==========
[sudo: authenticate] Password:
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 49408 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   123 KBytes  1.01 Mbits/sec  87
[  5]   1.00-2.00   sec   122 KBytes   996 Kbits/sec  86
[  5]   2.00-3.00   sec   123 KBytes  1.01 Mbits/sec  87
[  5]   3.00-4.00   sec   122 KBytes   997 Kbits/sec  86
[  5]   4.00-5.00   sec   122 KBytes   996 Kbits/sec  86
[  5]   5.00-6.00   sec   123 KBytes  1.01 Mbits/sec  87
[  5]   6.00-7.00   sec   122 KBytes   996 Kbits/sec  86
[  5]   7.00-8.00   sec   122 KBytes   996 Kbits/sec  86
[  5]   8.00-9.00   sec   123 KBytes  1.01 Mbits/sec  87
[  5]   9.00-10.00  sec   122 KBytes   993 Kbits/sec  86
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.00  sec  1.19 MBytes  1.00 Mbits/sec  0.000 ms  0/864 (0%)  sender
[  5]   0.00-10.01  sec  1.19 MBytes  1.00 Mbits/sec  0.146 ms  0/864 (0%)  receiver

Server output:
Accepted connection from 192.168.10.2, port 55040
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 49408
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   123 KBytes  1.01 Mbits/sec  0.039 ms  0/87 (0%)
[  5]   1.00-2.00   sec   122 KBytes   995 Kbits/sec  0.078 ms  0/86 (0%)
[  5]   2.00-3.00   sec   123 KBytes  1.01 Mbits/sec  0.094 ms  0/87 (0%)
[  5]   3.00-4.00   sec   122 KBytes   997 Kbits/sec  0.165 ms  0/86 (0%)
[  5]   4.00-5.00   sec   122 KBytes   995 Kbits/sec  0.090 ms  0/86 (0%)
[  5]   5.00-6.00   sec   123 KBytes  1.01 Mbits/sec  0.052 ms  0/87 (0%)
[  5]   6.00-7.00   sec   122 KBytes   996 Kbits/sec  0.033 ms  0/86 (0%)
[  5]   7.00-8.00   sec   122 KBytes   996 Kbits/sec  0.375 ms  0/86 (0%)
[  5]   8.00-9.00   sec   122 KBytes   997 Kbits/sec  0.036 ms  0/86 (0%)
[  5]   9.00-10.00  sec   123 KBytes  1.01 Mbits/sec  0.146 ms  0/87 (0%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.01  sec  1.19 MBytes  1.00 Mbits/sec  0.146 ms  0/864 (0%)  receiver


iperf Done.
========== TESTING 2 Mbps ==========
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 48996 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   245 KBytes  2.00 Mbits/sec  173
[  5]   1.00-2.00   sec   245 KBytes  2.00 Mbits/sec  173
[  5]   2.00-3.00   sec   245 KBytes  2.00 Mbits/sec  173
[  5]   3.00-4.00   sec   243 KBytes  1.99 Mbits/sec  172
[  5]   4.00-5.00   sec   245 KBytes  2.01 Mbits/sec  173
[  5]   5.00-6.00   sec   245 KBytes  2.00 Mbits/sec  173
[  5]   6.00-7.00   sec   243 KBytes  1.99 Mbits/sec  172
[  5]   7.00-8.00   sec   245 KBytes  2.00 Mbits/sec  173
[  5]   8.00-9.00   sec   243 KBytes  1.99 Mbits/sec  173
[  5]   9.00-10.00  sec   245 KBytes  2.00 Mbits/sec  172
```

Bandwidth tests: 2 and 4 Mbps:

```
[  5]   8.00-9.00   sec   243 KBytes  1.99 Mbits/sec  173
[  5]   9.00-10.00  sec   245 KBytes  2.00 Mbits/sec  172
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.00  sec  2.38 MBytes  2.00 Mbits/sec  0.000 ms  0/1727 (0%)  sender
[  5]   0.00-10.00  sec  2.38 MBytes  2.00 Mbits/sec  0.135 ms  0/1727 (0%)  receiver

Server output:
-----------------------------------------------------------
Server listening on 5201 (test #2)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 36620
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 48996
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   245 KBytes  2.00 Mbits/sec  0.073 ms  0/173 (0%)
[  5]   1.00-2.00   sec   245 KBytes  2.00 Mbits/sec  0.074 ms  0/173 (0%)
[  5]   2.00-3.00   sec   245 KBytes  2.00 Mbits/sec  0.083 ms  0/173 (0%)
[  5]   3.00-4.00   sec   243 KBytes  1.99 Mbits/sec  0.271 ms  0/172 (0%)
[  5]   4.00-5.00   sec   245 KBytes  2.00 Mbits/sec  0.053 ms  0/173 (0%)
[  5]   5.00-6.00   sec   243 KBytes  1.99 Mbits/sec  0.039 ms  0/172 (0%)
[  5]   6.00-7.00   sec   245 KBytes  2.00 Mbits/sec  0.256 ms  0/173 (0%)
[  5]   7.00-8.00   sec   245 KBytes  2.00 Mbits/sec  0.051 ms  0/173 (0%)
[  5]   8.00-9.00   sec   243 KBytes  1.99 Mbits/sec  0.114 ms  0/172 (0%)
[  5]   9.00-10.00  sec   245 KBytes  2.00 Mbits/sec  0.135 ms  0/173 (0%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.00  sec  2.38 MBytes  2.00 Mbits/sec  0.135 ms  0/1727 (0%)  receiver


iperf Done.
========= TESTING 4 Mbps ==========
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 60082 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   489 KBytes  4.00 Mbits/sec  346
[  5]   1.00-2.00   sec   488 KBytes  4.00 Mbits/sec  345
[  5]   2.00-3.00   sec   489 KBytes  4.01 Mbits/sec  346
[  5]   3.00-4.00   sec   488 KBytes  4.00 Mbits/sec  345
[  5]   4.00-5.00   sec   488 KBytes  4.00 Mbits/sec  345
[  5]   5.00-6.00   sec   489 KBytes  4.01 Mbits/sec  346
[  5]   6.00-7.00   sec   488 KBytes  4.00 Mbits/sec  345
[  5]   7.00-8.00   sec   488 KBytes  4.00 Mbits/sec  345
[  5]   8.00-9.00   sec   489 KBytes  4.01 Mbits/sec  346
[  5]   9.00-10.00  sec   488 KBytes  3.99 Mbits/sec  345
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.00  sec  4.77 MBytes  4.00 Mbits/sec  0.000 ms  0/3454 (0%)  sender
[  5]   0.00-10.00  sec  4.77 MBytes  4.00 Mbits/sec  0.073 ms  0/3454 (0%)  receiver

Server output:
-----------------------------------------------------------
Server listening on 5201 (test #3)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 51054
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 60082
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   489 KBytes  4.00 Mbits/sec  0.038 ms  0/346 (0%)
[  5]   1.00-2.00   sec   488 KBytes  4.00 Mbits/sec  0.104 ms  0/345 (0%)
[  5]   2.00-3.00   sec   488 KBytes  4.00 Mbits/sec  0.075 ms  0/345 (0%)
[  5]   3.00-4.00   sec   489 KBytes  4.01 Mbits/sec  0.057 ms  0/346 (0%)
[  5]   4.00-5.00   sec   488 KBytes  4.00 Mbits/sec  0.031 ms  0/345 (0%)
[  5]   5.00-6.00   sec   489 KBytes  4.01 Mbits/sec  0.057 ms  0/346 (0%)
[  5]   6.00-7.00   sec   488 KBytes  4.00 Mbits/sec  0.105 ms  0/345 (0%)
[  5]   7.00-8.00   sec   488 KBytes  3.99 Mbits/sec  0.088 ms  0/345 (0%)
[  5]   8.00-9.00   sec   489 KBytes  4.01 Mbits/sec  0.063 ms  0/346 (0%)
[  5]   9.00-10.00  sec   488 KBytes  4.00 Mbits/sec  0.073 ms  0/345 (0%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
```

Bandwidth tests: 5, 7 and 10 Mbps:

```
[  5]   0.00-1.00   sec   595 KBytes  4.87 Mbits/sec  0.830 ms  407/481 (8.7%)
[  5]   1.00-2.00   sec   592 KBytes  4.85 Mbits/sec  0.722 ms  185/604 (31%)
[  5]   2.00-3.00   sec   594 KBytes  4.87 Mbits/sec  0.818 ms  186/606 (31%)
[  5]   3.00-4.00   sec   592 KBytes  4.85 Mbits/sec  0.979 ms  184/603 (31%)
[  5]   4.00-5.00   sec   594 KBytes  4.86 Mbits/sec  0.872 ms  185/605 (31%)
[  5]   5.00-6.00   sec   592 KBytes  4.85 Mbits/sec  0.876 ms  185/604 (31%)
[  5]   6.00-7.00   sec   592 KBytes  4.86 Mbits/sec  0.939 ms  185/604 (31%)
[  5]   7.00-8.00   sec   594 KBytes  4.87 Mbits/sec  0.840 ms  185/605 (31%)
[  5]   8.00-9.00   sec   591 KBytes  4.84 Mbits/sec  0.836 ms  185/603 (31%)
[  5]   9.00-10.00  sec   594 KBytes  4.87 Mbits/sec  0.728 ms  185/605 (31%)
[  5]  10.00-10.24  sec   141 KBytes  4.81 Mbits/sec  0.829 ms  44/144 (31%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.24  sec  5.93 MBytes  4.86 Mbits/sec  0.829 ms  1749/6044 (29%)  receiver


iperf Done.
========== TESTING 10 Mbps ==========
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 60226 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec  1.19 MBytes  10.0 Mbits/sec  865
[  5]   1.00-2.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   2.00-3.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   3.00-4.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   4.00-5.00   sec  1.19 MBytes  10.0 Mbits/sec  864
[  5]   5.00-6.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   6.00-7.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   7.00-8.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   8.00-9.00   sec  1.19 MBytes  9.99 Mbits/sec  863
[  5]   9.00-10.00  sec  1.19 MBytes  9.99 Mbits/sec  864
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.00  sec  11.9 MBytes  10.0 Mbits/sec  0.000 ms  0/8634 (0%)  sender
[  5]   0.00-10.24  sec  5.93 MBytes  4.86 Mbits/sec  0.509 ms  4337/8632 (50%)  receiver

Server output:
-----------------------------------------------------------
Server listening on 5201 (test #6)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 42698
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 60226
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   595 KBytes  4.87 Mbits/sec  0.396 ms  237/658 (36%)
[  5]   1.00-2.00   sec   594 KBytes  4.87 Mbits/sec  0.260 ms  444/864 (51%)
[  5]   2.00-3.00   sec   592 KBytes  4.85 Mbits/sec  0.212 ms  444/863 (51%)
[  5]   3.00-4.00   sec   592 KBytes  4.86 Mbits/sec  0.225 ms  443/862 (51%)
[  5]   4.00-5.00   sec   594 KBytes  4.87 Mbits/sec  0.329 ms  444/864 (51%)
[  5]   5.00-6.00   sec   592 KBytes  4.85 Mbits/sec  0.262 ms  444/863 (51%)
[  5]   6.00-7.00   sec   594 KBytes  4.87 Mbits/sec  0.259 ms  444/864 (51%)
[  5]   7.00-8.00   sec   592 KBytes  4.85 Mbits/sec  0.373 ms  445/864 (52%)
[  5]   8.00-9.00   sec   592 KBytes  4.85 Mbits/sec  0.429 ms  444/863 (51%)
[  5]   9.00-10.00  sec   594 KBytes  4.86 Mbits/sec  0.648 ms  444/864 (51%)
[  5]  10.00-10.24  sec   140 KBytes  4.80 Mbits/sec  0.509 ms  104/203 (51%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.24  sec  5.93 MBytes  4.86 Mbits/sec  0.509 ms  4337/8632 (50%)  receiver


iperf Done.
========== ROUTER QUEUE STATISTICS ==========
qdisc htb 1: root refcnt 13 r2q 10 default 0x10 direct_packets_stat 0 direct_qlen 1000
 Sent 28212350 bytes 19029 pkt (dropped 6113, overlimits 12999 requeues 0)
 backlog 0b 0p requeues 0
qdisc pfifo 10: parent 1:10 limit 100p
 Sent 28212350 bytes 19029 pkt (dropped 6113, overlimits 0 requeues 0)
 backlog 0b 0p requeues 0
naman@DESKTOP-PQFNDR1:~$
```

Receiver server output and final results command:

```
Accepted connection from 192.168.10.2, port 42698
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 60226
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   595 KBytes  4.87 Mbits/sec  0.396 ms  237/658 (36%)
[  5]   1.00-2.00   sec   594 KBytes  4.87 Mbits/sec  0.260 ms  444/864 (51%)
[  5]   2.00-3.00   sec   592 KBytes  4.85 Mbits/sec  0.212 ms  444/863 (51%)
[  5]   3.00-4.00   sec   592 KBytes  4.86 Mbits/sec  0.225 ms  443/862 (51%)
[  5]   4.00-5.00   sec   594 KBytes  4.87 Mbits/sec  0.329 ms  444/864 (51%)
[  5]   5.00-6.00   sec   592 KBytes  4.85 Mbits/sec  0.262 ms  444/863 (51%)
[  5]   6.00-7.00   sec   594 KBytes  4.87 Mbits/sec  0.259 ms  444/864 (51%)
[  5]   7.00-8.00   sec   592 KBytes  4.85 Mbits/sec  0.373 ms  445/864 (52%)
[  5]   8.00-9.00   sec   592 KBytes  4.85 Mbits/sec  0.429 ms  444/863 (51%)
[  5]   9.00-10.00  sec   594 KBytes  4.86 Mbits/sec  0.648 ms  444/864 (51%)
[  5]  10.00-10.24  sec   140 KBytes  4.80 Mbits/sec  0.509 ms  104/203 (51%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.24  sec  5.93 MBytes  4.86 Mbits/sec  0.509 ms  4337/8632 (50%)  receiver
-----------------------------------------------------------
Server listening on 5201 (test #7)
-----------------------------------------------------------
echo "========== SKILL 3 FINAL RESULTS =========="

for rate in 1 2 4 5 7 10; do
    echo ""
    echo "========== ${rate} Mbps =========="
    grep -E "receiver$" ~/skill3_results/rate_${rate}Mbps.txt | tail -1
done

echo ""
echo "========== FINAL QUEUE STATISTICS =========="
sudo ip netns exec router tc -s qdisc show dev veth-r2
```

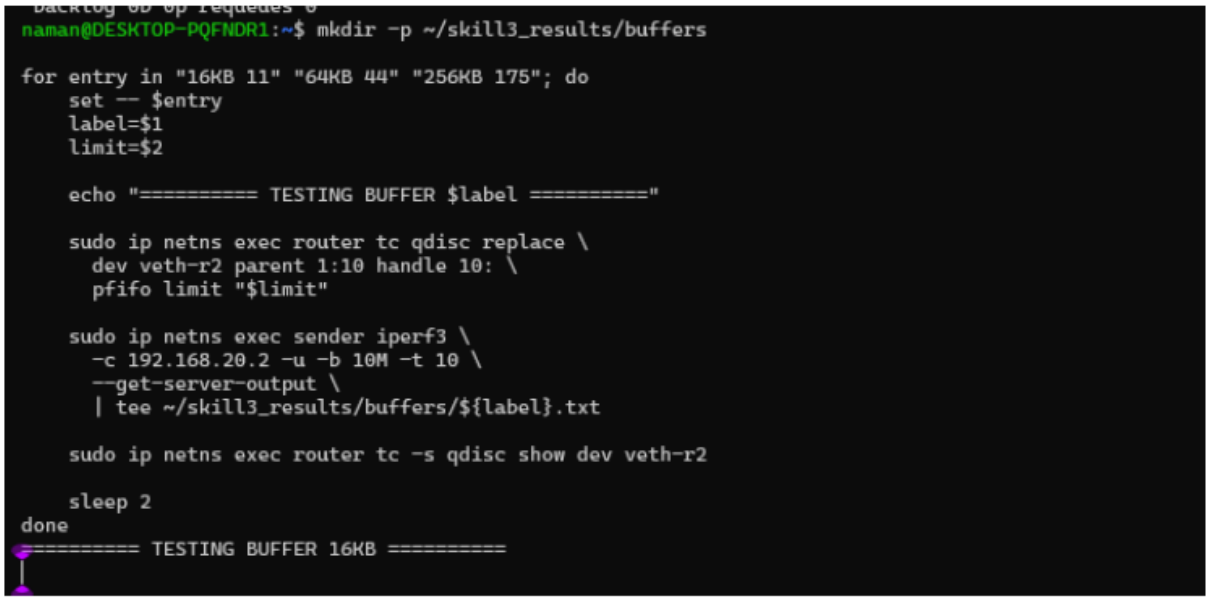
measured throughput, loss and queue statistics:

```
 backlog 0b 0p requeues 0
naman@DESKTOP-PQFNDR1:~$ echo "========== SKILL 3 FINAL RESULTS =========="

for rate in 1 2 4 5 7 10; do
    echo "========== ${rate} Mbps =========="
    grep "receiver$" ~/skill3_results/rate_${rate}Mbps.txt | tail -1
done

echo "========== FINAL QUEUE STATISTICS =========="
sudo ip netns exec router tc -s qdisc show dev veth-r2
========== SKILL 3 FINAL RESULTS ==========
========== 1 Mbps ==========
[  5]   0.00-10.01  sec  1.19 MBytes  1.00 Mbits/sec  0.146 ms  0/864 (0%)  receiver
========== 2 Mbps ==========
[  5]   0.00-10.00  sec  2.38 MBytes  2.00 Mbits/sec  0.135 ms  0/1727 (0%)  receiver
========== 4 Mbps ==========
[  5]   0.00-10.00  sec  4.77 MBytes  4.00 Mbits/sec  0.073 ms  0/3454 (0%)  receiver
========== 5 Mbps ==========
[  5]   0.00-10.24  sec  5.93 MBytes  4.86 Mbits/sec  0.541 ms  24/4317 (0.56%)  receiver
========== 7 Mbps ==========
[  5]   0.00-10.24  sec  5.93 MBytes  4.86 Mbits/sec  0.829 ms  1749/6044 (29%)  receiver
========== 10 Mbps ==========
[  5]   0.00-10.24  sec  5.93 MBytes  4.86 Mbits/sec  0.509 ms  4337/8632 (50%)  receiver
========== FINAL QUEUE STATISTICS ==========
qdisc htb 1: root refcnt 13 r2q 10 default 0x10 direct_packets_stat 0 direct_qlen 1000
 Sent 28212420 bytes 19030 pkt (dropped 6113, overlimits 12999 requeues 0)
 backlog 0b 0p requeues 0
qdisc pfifo 10: parent 1:10 limit 100p
 Sent 28212420 bytes 19030 pkt (dropped 6113, overlimits 0 requeues 0)
 backlog 0b 0p requeues 0
naman@DESKTOP-PQFNDR1:~$
```

Buffer test commands for 16 KB, 64 KB and 256 KB:



Buffer test: 16 KB result:

```
naman@DESKTOP-PQFNDR1:~$ mkdir -p ~/skill3_results/buffers

for entry in "16KB 11" "64KB 44" "256KB 175"; do
    set -- $entry
    label=$1
    limit=$2

    echo "========== TESTING BUFFER $label =========="

    sudo ip netns exec router tc qdisc replace \
      dev veth-r2 parent 1:10 handle 10: \
      pfifo limit "$limit"

    sudo ip netns exec sender iperf3 \
      -c 192.168.20.2 -u -b 10M -t 10 \
      --get-server-output \
      | tee ~/skill3_results/buffers/${label}.txt

    sudo ip netns exec router tc -s qdisc show dev veth-r2

    sleep 2
done
========== TESTING BUFFER 16KB ==========
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 49694 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec  1.19 MBytes  10.0 Mbits/sec  865
[  5]   1.00-2.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   2.00-3.00   sec  1.19 MBytes  9.99 Mbits/sec  862
[  5]   3.00-4.00   sec  1.19 MBytes  10.0 Mbits/sec  864
[  5]   4.00-5.00   sec  1.19 MBytes  10.0 Mbits/sec  864
[  5]   5.00-6.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   6.00-7.00   sec  1.19 MBytes  9.99 Mbits/sec  864
[  5]   7.00-8.00   sec  1.19 MBytes  9.99 Mbits/sec  862
[  5]   8.00-9.00   sec  1.19 MBytes  10.0 Mbits/sec  864
[  5]   9.00-10.00  sec  1.19 MBytes  9.98 Mbits/sec  863
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.00  sec  11.9 MBytes  10.0 Mbits/sec  0.000 ms  0/8634 (0%)  sender
[  5]   0.00-10.03  sec  5.81 MBytes  4.86 Mbits/sec  0.345 ms  4428/8634 (51%)  receiver

Server output:
-----------------------------------------------------------
Server listening on 5201 (test #7)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 42514
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 49694
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   595 KBytes  4.87 Mbits/sec  0.781 ms  419/840 (50%)
[  5]   1.00-2.00   sec   592 KBytes  4.85 Mbits/sec  0.136 ms  445/864 (52%)
[  5]   2.00-3.00   sec   594 KBytes  4.86 Mbits/sec  0.285 ms  444/864 (51%)
[  5]   3.00-4.00   sec   592 KBytes  4.85 Mbits/sec  0.309 ms  444/863 (51%)
[  5]   4.00-5.00   sec   594 KBytes  4.87 Mbits/sec  0.478 ms  443/863 (51%)
[  5]   5.00-6.00   sec   592 KBytes  4.85 Mbits/sec  0.300 ms  445/864 (52%)
[  5]   6.00-7.00   sec   594 KBytes  4.87 Mbits/sec  0.299 ms  444/864 (51%)
[  5]   7.00-8.00   sec   592 KBytes  4.85 Mbits/sec  0.172 ms  443/862 (51%)
[  5]   8.00-9.00   sec   594 KBytes  4.87 Mbits/sec  0.347 ms  445/865 (51%)
[  5]   9.00-10.00  sec   591 KBytes  4.84 Mbits/sec  0.391 ms  444/862 (52%)
[  5]  10.00-10.03  sec  15.6 KBytes  4.64 Mbits/sec  0.345 ms  12/23 (52%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.03  sec  5.81 MBytes  4.86 Mbits/sec  0.345 ms  4428/8634 (51%)  receiver


iperf Done.
qdisc htb 1: root refcnt 13 r2q 10 default 0x10 direct_packets_stat 0 direct_qlen 1000
 Sent 34481145 bytes 23256 pkt (dropped 10551, overlimits 17200 requeues 0)
 backlog 0b 0p requeues 0
```

Buffer test: 64 KB result:

```
qdisc pfifo 10: parent 1:10 limit 11p
 Sent 34481145 bytes 23256 pkt (dropped 10551, overlimits 0 requeues 0)
 backlog 0b 0p requeues 0
========== TESTING BUFFER 64KB ==========
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 45083 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec  1.19 MBytes  10.0 Mbits/sec  865
[  5]   1.00-2.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   2.00-3.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   3.00-4.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   4.00-5.00   sec  1.19 MBytes  10.0 Mbits/sec  865
[  5]   5.00-6.00   sec  1.19 MBytes  9.99 Mbits/sec  862
[  5]   6.00-7.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   7.00-8.01   sec  1.19 MBytes  9.98 Mbits/sec  866
[  5]   8.01-9.00   sec  1.19 MBytes  10.0 Mbits/sec  860
[  5]   9.00-10.00  sec  1.19 MBytes  10.0 Mbits/sec  864
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.00  sec  11.9 MBytes  10.0 Mbits/sec  0.000 ms  0/8634 (0%)  sender
[  5]   0.00-10.11  sec  5.85 MBytes  4.86 Mbits/sec  0.296 ms  4397/8633 (51%)  receiver

Server output:
-----------------------------------------------------------
Server listening on 5201 (test #8)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 46184
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 45083
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   592 KBytes  4.85 Mbits/sec  0.507 ms  353/772 (46%)
[  5]   1.00-2.00   sec   592 KBytes  4.85 Mbits/sec  0.228 ms  445/864 (52%)
[  5]   2.00-3.00   sec   594 KBytes  4.86 Mbits/sec  0.417 ms  445/865 (51%)
[  5]   3.00-4.00   sec   592 KBytes  4.85 Mbits/sec  0.257 ms  443/862 (51%)
[  5]   4.00-5.00   sec   594 KBytes  4.86 Mbits/sec  0.224 ms  444/864 (51%)
[  5]   5.00-6.00   sec   592 KBytes  4.86 Mbits/sec  0.261 ms  445/864 (52%)
[  5]   6.00-7.00   sec   591 KBytes  4.84 Mbits/sec  0.253 ms  444/862 (52%)
[  5]   7.00-8.01   sec   597 KBytes  4.87 Mbits/sec  0.550 ms  446/868 (51%)
[  5]   8.01-9.00   sec   590 KBytes  4.85 Mbits/sec  0.313 ms  442/859 (51%)
[  5]   9.00-10.00  sec   592 KBytes  4.85 Mbits/sec  0.169 ms  443/862 (51%)
[  5]  10.00-10.11  sec  62.2 KBytes  4.83 Mbits/sec  0.296 ms  47/91 (52%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.11  sec  5.85 MBytes  4.86 Mbits/sec  0.296 ms  4397/8633 (51%)  receiver


iperf Done.
qdisc htb 1: root refcnt 13 r2q 10 default 0x10 direct_packets_stat 0 direct_qlen 1000
 Sent 40794277 bytes 27507 pkt (dropped 14949, overlimits 21422 requeues 0)
 backlog 0b 0p requeues 0
qdisc pfifo 10: parent 1:10 limit 44p
 Sent 40794277 bytes 27507 pkt (dropped 14949, overlimits 0 requeues 0)
 backlog 0b 0p requeues 0
========== TESTING BUFFER 256KB ==========
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 34412 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec  1.19 MBytes  10.0 Mbits/sec  864
[  5]   1.00-2.00   sec  1.19 MBytes  10.0 Mbits/sec  864
[  5]   2.00-3.00   sec  1.19 MBytes  9.99 Mbits/sec  863
[  5]   3.00-4.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   4.00-5.00   sec  1.19 MBytes  9.99 Mbits/sec  864
[  5]   5.00-6.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   6.00-7.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   7.00-8.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   8.00-9.00   sec  1.19 MBytes  10.0 Mbits/sec  864
[  5]   9.00-10.00  sec  1.19 MBytes  9.98 Mbits/sec  863
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
```

Buffer test: 256 KB result:

```
========== TESTING BUFFER 256KB ==========
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 34412 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec  1.19 MBytes  10.0 Mbits/sec  864
[  5]   1.00-2.00   sec  1.19 MBytes  10.0 Mbits/sec  864
[  5]   2.00-3.00   sec  1.19 MBytes  9.99 Mbits/sec  863
[  5]   3.00-4.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   4.00-5.00   sec  1.19 MBytes  9.99 Mbits/sec  864
[  5]   5.00-6.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   6.00-7.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   7.00-8.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   8.00-9.00   sec  1.19 MBytes  10.0 Mbits/sec  864
[  5]   9.00-10.00  sec  1.19 MBytes  9.98 Mbits/sec  863
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.00  sec  11.9 MBytes  10.0 Mbits/sec  0.000 ms  0/8634 (0%)  sender
[  5]   0.00-10.42  sec  6.03 MBytes  4.86 Mbits/sec  0.349 ms  4263/8633 (49%)  receiver

Server output:
-----------------------------------------------------------
Server listening on 5201 (test #9)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 45650
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 34412
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   594 KBytes  4.86 Mbits/sec  0.223 ms  83/503 (17%)
[  5]   1.00-2.00   sec   594 KBytes  4.86 Mbits/sec  0.341 ms  445/865 (51%)
[  5]   2.00-3.00   sec   592 KBytes  4.85 Mbits/sec  0.213 ms  443/862 (51%)
[  5]   3.00-4.00   sec   592 KBytes  4.86 Mbits/sec  0.271 ms  443/862 (51%)
[  5]   4.00-5.00   sec   594 KBytes  4.86 Mbits/sec  0.543 ms  446/866 (52%)
[  5]   5.00-6.00   sec   592 KBytes  4.86 Mbits/sec  0.456 ms  443/862 (51%)
[  5]   6.00-7.00   sec   594 KBytes  4.87 Mbits/sec  0.213 ms  444/864 (51%)
[  5]   7.00-8.00   sec   592 KBytes  4.85 Mbits/sec  0.237 ms  444/863 (51%)
[  5]   8.00-9.00   sec   594 KBytes  4.87 Mbits/sec  0.495 ms  444/864 (51%)
[  5]   9.00-10.00  sec   592 KBytes  4.85 Mbits/sec  0.251 ms  443/862 (51%)
[  5]  10.00-10.42  sec   247 KBytes  4.84 Mbits/sec  0.349 ms  185/360 (51%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.42  sec  6.03 MBytes  4.86 Mbits/sec  0.349 ms  4263/8633 (49%)  receiver


iperf Done.
qdisc htb 1: root refcnt 13 r2q 10 default 0x10 direct_packets_stat 0 direct_qlen 1000
 Sent 47307334 bytes 31896 pkt (dropped 19213, overlimits 25784 requeues 0)
 backlog 0b 0p requeues 0
qdisc pfifo 10: parent 1:10 limit 175p
 Sent 47307334 bytes 31896 pkt (dropped 19213, overlimits 0 requeues 0)
 backlog 0b 0p requeues 0
naman@DESKTOP-PQFNDR1:~$
```

Observation and Record — Skill 3

1. Experiment setup record

- Topology: Sender (192.168.10.2) → Router (192.168.10.1 / 192.168.20.1) → Receiver (192.168.20.2).
- Router outgoing interface: veth-r2. Configured bottleneck: 5 Mbit/s using HTB. Queue discipline: pfifo.
- Traffic generation: iperf3 UDP tests, 10 seconds per offered load.
- Connectivity verification: 4 packets sent, 4 received, 0% loss before load testing.

2. Observation table: offered load versus throughput

Offered load (Mbps)	Receiver throughput (Mbps)	UDP loss	Jitter (ms)	Observation
1	1.00	0%	0.146	Below capacity; no loss
2	2.00	0%	0.135	Below capacity; no loss
4	4.00	0%	0.073	Below capacity; no loss
5	4.86	0.56%	0.541	Near capacity
7	4.86	29%	0.829	Congestion and loss
10	4.86	50%	0.509	Severe congestion and loss

Record: Throughput increased with offered load until approximately 5 Mbps, then stayed near 4.86 Mbps. Loss increased from 0% at 1–4 Mbps to 29% at 7 Mbps and 50% at 10 Mbps.

3. Observation table: buffer size comparison at 10 Mbps

Nominal buffer	Actual pfifo limit	Throughput	Packet loss	Jitter
16 KB	11 packets	4.86 Mbps	51%	0.345 ms
64 KB	44 packets	4.86 Mbps	51%	0.296 ms
256 KB	175 packets	4.86 Mbps	49%	0.349 ms

Record: Larger buffer limits did not increase the receiver throughput. Loss was 51%, 51%, and 49% in the three runs. The queue limits are packet counts approximating the named buffer sizes, not byte-exact limits.

4. Queue statistics record

After the offered-load series, the router showed 6,113 dropped packets and 12,999 overlimits. After the three buffer tests, the cumulative router queue statistics showed 19,213 dropped packets and 25,784 overlimits. These are cumulative counters; they must not be treated as drops from a single test.

5. Little's Law and queue-delay calculation

Little's Law: $L = \lambda \times W$, so $W = L / \lambda$. Here L is the average queue occupancy, λ is the average admitted packet rate, and W is average waiting time.

The following are theoretical full-buffer serialization times, calculated as (buffer size in bits) / 5,000,000 bits per second. They are NOT observed average queueing delays.

Nominal size (binary units)	Calculation	Theoretical time
16 KiB	16 × 1024 × 8 / 5,000,000	26.2 ms
64 KiB	64 × 1024 × 8 / 5,000,000	104.9 ms
256 KiB	256 × 1024 × 8 / 5,000,000	419.4 ms

Measured average queue occupancy and per-packet queueing delay were not collected in the supplied evidence. Therefore Little's Law cannot be experimentally validated from these screenshots alone.

6. Recorded answers to observation questions

What happens when offered load increases?

Receiver throughput initially increases; after the 5 Mbps bottleneck is reached, throughput levels off near 4.86 Mbps and packet loss increases.

Where is the knee point?

The observed transition is near the 5 Mbps bottleneck. A precise delay-based knee point cannot be determined because queueing-delay measurements were not captured.

What is the effect of changing the buffer size?

At 10 Mbps offered load, throughput remained near 4.86 Mbps for all three buffers. Loss was 51%, 51%, and 49%, respectively.

Does a larger buffer always improve performance?

No. It can absorb bursts, but under sustained overload it may increase waiting time without improving throughput.

Can measured delay and Little's Law be compared?

Not quantitatively with the current screenshots. The report provides theoretical full-buffer times, but average queue occupancy and measured queueing delay are missing.

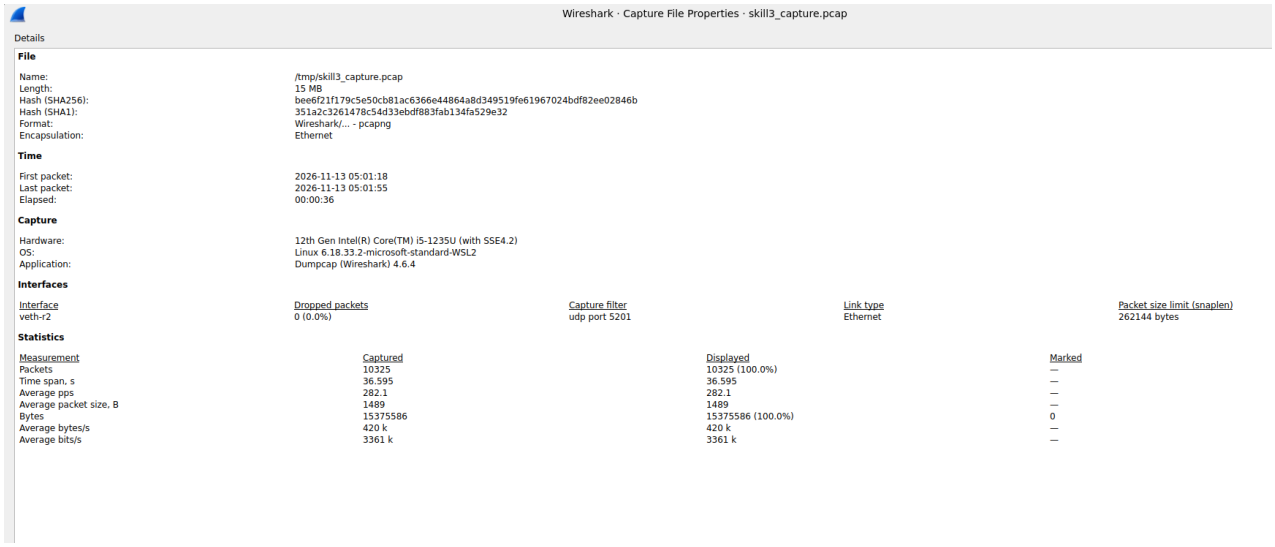
What do the results show about congestion?

When traffic exceeds the 5 Mbps router limit, extra packets are dropped while successful throughput remains close to the link capacity.

7. Final observation and conclusion

Wireshark I/O Graph

Capture File Properties



Wireshark · Capture File Properties · skill3_capture.pcap

Details

File

Name:	/tmp/skill3_capture.pcap
Length:	15 MB
Hash (SHA256):	bee6f21f179c5e50cb81ac6366e44864a8d349519fe61967024bdf82ee02846b
Hash (SHA1):	351a2c3261478c54d33ebdf883fab134fa529e32
Format:	Wireshark/... - pcapng
Encapsulation:	Ethernet

Time

First packet:	2026-11-13 05:01:18
Last packet:	2026-11-13 05:01:55
Elapsed:	00:00:36

Capture

Hardware:	12th Gen Intel(R) Core(TM) i5-1235U (with SSE4.2)
OS:	Linux 6.18.33.2-microsoft-standard-WSL2
Application:	Dumpcap (Wireshark) 4.6.4

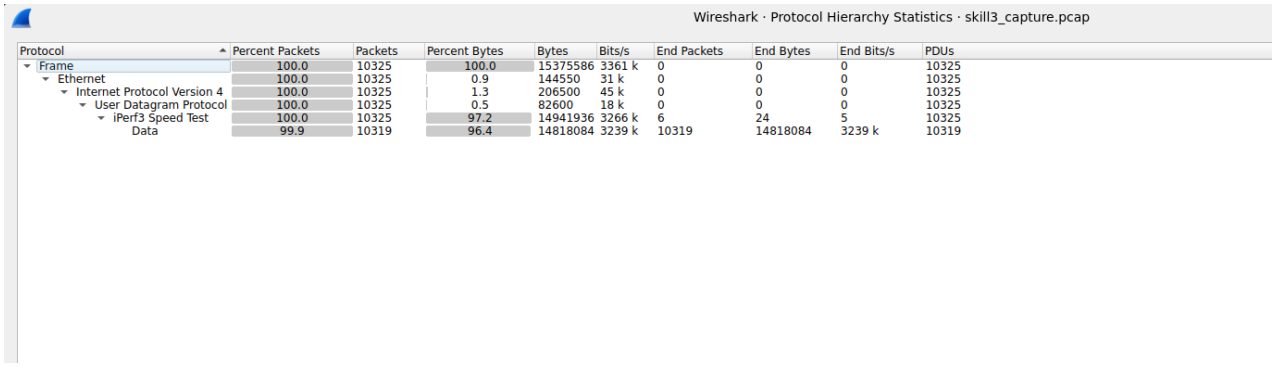
Interfaces

Interface	Dropped packets	Capture filter	Link type	Packet size limit (snaplen)
veth-r2	0 (0.0%)	udp port 5201	Ethernet	262144 bytes

Statistics

Measurement	Captured	Displayed	Marked
Packets	10325	10325 (100.0%)	—
Time span, s	36.595	36.595	—
Average pps	282.1	282.1	—
Average packet size, B	1489	1489	—
Bytes	15375586	15375586 (100.0%)	0
Average bytes/s	420 k	420 k	—
Average bits/s	3361 k	3361 k	—

Protocol Hierarchy



Wireshark · Protocol Hierarchy Statistics · skill3_capture.pcap

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	10325	100.0	15375586	3361 k	0	0	0	10325
Ethernet	100.0	10325	0.9	144550	31 k	0	0	0	10325
Internet Protocol Version 4	100.0	10325	1.3	206500	45 k	0	0	0	10325
User Datagram Protocol	100.0	10325	0.5	82600	18 k	0	0	0	10325
iPerf3 Speed Test	100.0	10325	97.2	14941936	3266 k	6	24	5	10325
Data	99.9	10319	96.4	14818084	3239 k	10319	14818084	3239 k	10319